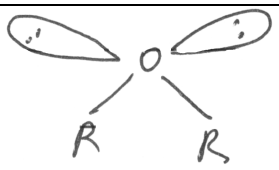
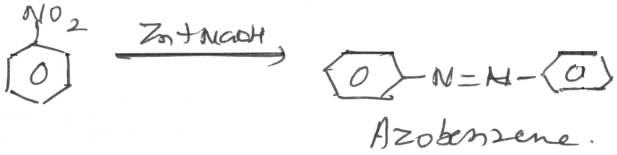
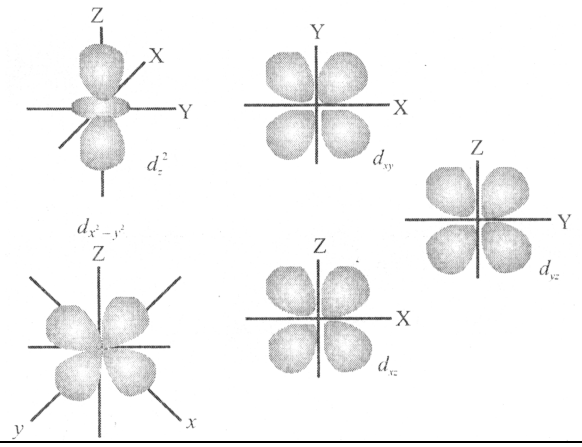
CODE: 67/ESS

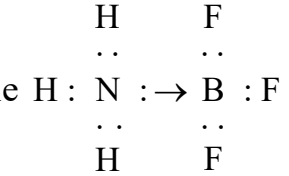
MARKING SCHEME
SENIOR SECONDARY
MARCH-APRIL 2024
CHEMISTRY
(313)

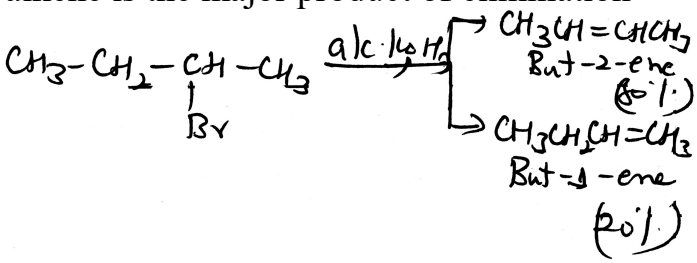
SET			EXPECTED VALUE POINTS FOR EACH STEP	Distribution Of Marks	Total Marks
A	B	C			
Q. No.					
			SECTION A		
1	-	-	B	1	1
2	-	-	B	1	1
3	-	-	(i) D OR (ii) A	1	1
4	-	-	(i) C OR (ii) B	1	1
5	-	-	(i) D OR (ii) D	1	1
6	-	-	(i) C OR (ii) A	1	1
7	-	-	(i) B OR (ii) D	1	1
8	-	-	B	1	1
9	-	-	(i)C OR (ii)C	1	1
10	-	-	C	1	1
11	-	-	(i) B OR (ii) D	1	1
12	-	-	A	1	1
13	-	-	(i) B OR (ii) D	1	1
14	-	-	D	1	1
15	-	-	D	1	1
16	-	-	A	1	1
17	-	-	(1) Molar Mass (2) C-12 (3) 0.001 (4) Integral	1×2	2
18	-	-	(1) Both bonding and non-bonding (2) Stability of the molecule (3) Trigonal bipyramidal (4) Non-bonding electrons	1×2	2

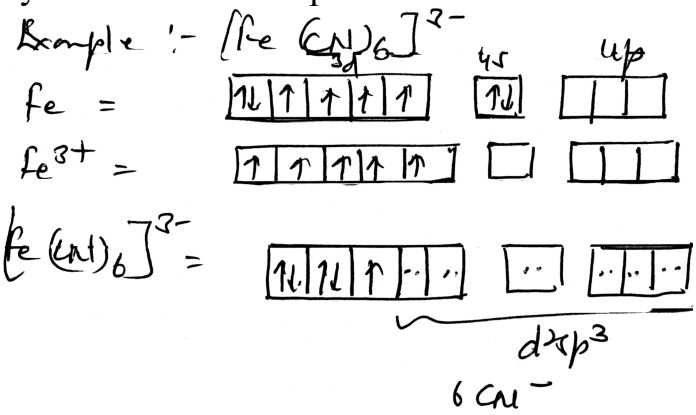
19	-	-	(1) True (2) False (3) False (4) True	1×2	2
20	-	-	(1) False (2) True (3) True (4) True	1×2	2
21	-	-	(1) True (2) False (3) True (4) True	1×2	2
22	-	-	(1) True (2) True (3) True (4) False	1×2	2
23	-	-	(1) True (2) False (3) True (4) True	1×2	2
24	-	-	1. $[Co(CN)_6]^{3-}$ is on inner orbital complex with d^2sp^3 hybridization while $[CoF_6]^{3-}$ is on outer orbital complex with sp^3d^2 hybridization. 2. Only $[Co(CH_3)_5\ Br]\ SO_4$ give white ppt. with $BaCl_2$ solution / only $[CO(NH_3)_5SO_4]Br$ give yellow ppt. with $AgNO_3$ solution. 3. EDTA 4. Ethylenediamine	1×2	2

25	-	-	1.  2. $CH_3OCH_3 + HBr \rightleftharpoons CH_3\overset{+}{\underset{H}{O}}CH_3Br^-$ (Or any other) 3. Methoxy propane 4. $ROH + Na \rightarrow \underset{\text{Metal alloxide}}{RO^-Na^+} + \frac{1}{2}H_2$ $RO^-Na^+ + R'\text{-}X \rightarrow R\text{-}O\text{-}R' + NaX$ (or any other)	1×2	2
26	-	-	1. 3-Methyl, 2- Nitro pentane 2. $CH_3CH_2CH_3 \xrightarrow[680k]{NaCO_3} CH_3CH_2CH_2NO_2 +$ $CH_3\ \underset{\underset{NO_2}{\vert}}{C}HCH_3 + CH_3NO_2 + CH_3CH_2NO_2$ 3.  Azobenzene. 4. Nitro compounds possess higher boiling points than the corresponding alkanes because of their polar nature.	1×2	2
27	-	-	(1) Progesterene – It prepares the lining of the uterus for the implantation of the fertilized ovum. (2) Diabetes mellitus; pancreas (3) (i) Riboflovin (Vitamin B_2) (ii) Pyridoxine (Vitamin B_6) (4) Vitamin A, D, E and K (any two)	1×2	2

28	-	-	(1) Ethylene glycol, terephthalic acid (2) Buna-S – formed from more than one kind of monomers. (3) Condensation polymerization (4) (a) Linear (b) Bronched chain (c) Cross linked	1x2	2
			SECTION B		
29	-	-	Mass of 1 mol of Na_3PO_4 = 3x (mass of 1 mol of Na) + mass of 1 mol of P + 4x (mass of 1 mol of oxygen) = 3×23.0 g + 31.0 g 4(16.0)g = 164.0 g mol^{-1}. Mass of 0.252 mol of Na_3PO_4 = 0.252×164.0 g = 41.328 g **OR** Percentage of carbon = 51.3% Percentage of oxygen = 46.9% Moles of carbon is 100 g of substance $= \frac{53.1}{12.0}\text{ g} = 4.43\text{ mol}$ Mol of oxygen $\frac{46.0}{16.0} = 2.93\text{ mol}$ Molar ratio C and O = $\frac{4.43}{2.93} : \frac{2.93}{2.93} = 1.50:1 \text{ or } 3:2$ Empirical formula = C_3O_2.	1×2	2
30	-	-		2	2

31	-	-	A band in which both the bending electrons are contributed by one atom only Example  Or any other **OR** Bond angle – It is the angle between the two bonds in a molecule (a) 107' 3° (b) 109° 28'	1 1 1 ½ + ½	2
32	-	-	Exothermic reactions are those reactions which proceed with the evolution of heat. Example – 2 cm^3 of dil. HCl added in a test tube containing a few pieces of granulated zinc. The test tube is hot. Or any other examples	1 1	2
33	-	-	Ionic product – when water molecules undergo auto ionization, half of them acting as an acid and half of them acting as a base. $H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$ $K_{eq} = \frac{[H_3O^+][OH^-]}{[H_2O]^2}$ $[H_3O^+][OH^-] = K_{eq} \times [H_2O]^2 = K_w$	1 1	2
34	-	-	Copper is discharged at Cathode and $O_2(g)$ at anode. At Anode $\Rightarrow 2H_2O \longrightarrow 4H$	1x 2	2
35	-	-	$XeF_6 + H_2O \rightarrow XeOF_4 + 2HF$ $XeF_6 + 2H_2O \rightarrow XeO_2F_2 + 4HF$	1 1	2

36	-	-	Step 1: Conc. H_2SO_4 is added to sodium chromate. $2Na_2CrO_4 + H_2SO_4 \rightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$ Step 2: Saturated solution of KCl is added to it. $Na_2CrO_7 + 2KCl \rightarrow K_2Cr_2O_7 + 2NaCl$	1+1	2
37	-	-	Saytzeff's rule – when the elimination can happen from an alkyl halide in two different ways. Then the more highly substituted alkene is the major product of elimination  (of any other example)	1 1	2
38	-	-	Methane – In methane the central carbon atom would have four pairs of electrons in its valence shell. According to VSEPR theory, these would be placed tetrahedrally around the carbon atom. So, CH_4 has a tetrahedral shape. Postulate – The electron pairs around the control atom in a molecule arrange themselves in space in such a way that they minimize their mutual repulsion.	1 1 1	3
39	-	-	(i) Here q = + 15KJ, w = -5kJ $\Delta u = q+w = 15 + (-5) = 10$ kJ. The internal energy of the system increases by 10 kJ (ii) w = +5 kJ, q = –15 kJ, $\Delta u = q+w = -15 + (+5) = -10$ kJ The internal energy of the system decreases by 10 kJ.	½ ½ ½ ½ ½ ½	3

40	-	-	(i)(a) $[Co(H_2O)_6]Cl_3$ (b) Inner orbital complex is that complex in which inner d-orbitals are t involved in hybridization i.e. d^2sp^3.  It is an inner orbital complex. Or any other example **OR** (ii) In $[Fe(CN)_6]^{3-}$, CN is a strong ligand. Hence electrons pair up in the d-subshell and there is only unpaired electron. In $[FeF_6]^{3-}$, F^- is a weak ligand. Electrons do not pair up. There are five unpaired d-electrons. Therefore $[Fe\ F_6]^{3-}$ has greater magnetic moment than $[Fe\ (CN)_6]^{3-}$.	1 ½ ½ ½ ½ 3	3

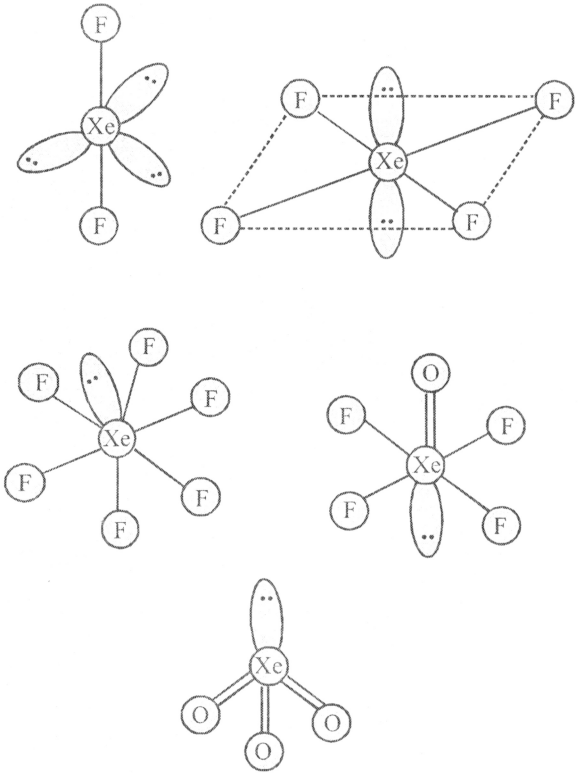
41	-	-	(i) (a) Because they have one alkyl group whereas a keytone has two alkyl groups. Hence carbonyl group in aldehydes is more positive. (b) $CH_3CH_2MgBr + CH_3-\overset{O}{\overset{\|\|}{C}}-CH_3 \rightarrow CH_3-\underset{CH_2CH_3}{\overset{CH_3}{\overset{\|}{\underset{\|}{C}}}}-OH$ (Tertiary alcohol) (c) $\underset{\text{Butonoic acid}}{CH_3CH_2CH_2COOH} \xrightarrow[(2)\ H_2O]{(1)\ Br/p} \underset{2-\text{Bromobutanoic acid}}{CH_3CH_2\overset{Br}{\overset{\|}{C}}HCHCOOH}$ **OR** (ii) (a) 2-chlorobutonic acid because of maximum -I effect of –Cl at 2nd position. (b) Tollen's test = Tollen's reagent – $\underset{\text{Aldehyde}}{R-\overset{O}{\overset{\|\|}{C}}-H} \xrightarrow[H_2O]{\left[Ag(NH_3)_2\right]^+} R-\overset{O}{\overset{\|\|}{C}}-O^- + Ag \rightarrow \text{mirror}$ (c) By heating in a basic solution of hydrazine in a high boiling alcohol/chemical equator OR by using amalgamated zinc and hydrochloric acid / chemical equation involved.	1 1 1 1 1 1	3

42	-	-	(a) It is the depression in freezing point of a solution when a mole of a solute is dissolved in 1 kg of the solvent. $AT_f = K_f m$ (b) $K_f = 1.86$ k/m $\text{Molality of glucose} = \frac{\text{wt. of glucose}}{\text{Mol. wt.}} \times \frac{1000}{\text{wt. of solvent}}$ $m = \frac{0.52}{180} \times \frac{1000}{80.2} = 0.036$ $\Delta T_f = 1.86 \times 0.036 = 0.66\ k$ $\therefore \text{freezing point} = 273 - 0.66\ k$ $= 272.34\ k$ **OR** (a) When the molecular mass of a substance determined by studying any of the colligative properties comes out to be different then the theoretically expected value, the substance is said to show abnormal molecular mass. $i = \frac{\text{calculated molecular mass}}{\text{observed molecular mass}}$ (b) $\pi = iCRT = i\frac{n}{V}RT$ $n = \frac{\pi \times V}{i \times R \times T}$ $= \frac{0.75\ \text{atm} \times 2.5\ L}{2.47 \times 0.0821\ \text{at m k}^{-1}\ \text{mol}^{-1} \times 200\ k}$ $= 0.0308$ mole molar mass of $CaCl_2 = 111\ g\ mol^{-1}$ Amount dissolved $= 0.0308 \times 111\ g$ $= 3.42\ g$	1 1 ½ 1 ½ 1 1 1 ½ 1 ½ 1	5

43	-	-	(a) $SnCl_4 < GeCl_4 < SiCl_4 < CCl_4$ (b) The bind angles decreases from 107° to 90°. (c) The covalent characters of a halide increases with increase in the oxidation state of the element forming halides. (d) (i) $4Al + 3O_2 \rightarrow 2Al_2O_3$ (ii) $C + 2Cl \rightarrow CCl_4$ **OR** (a) In increases because of the decrease in atomic size along a period. The larger the atom, the less strongly the electrons are held by the nucleus. (b) (i) Acidic nature increases with increase in oxidation state of the element. Basic nature increases with increase in atomic number. (ii) Ionic nature increases from the group. Covalent nature decreases down the group. (c) Na < Be < N < He	1 1 1 1 1 2 2 1	5

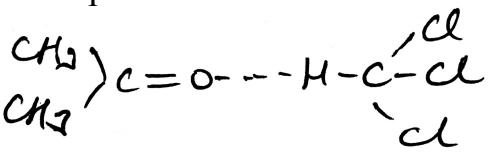
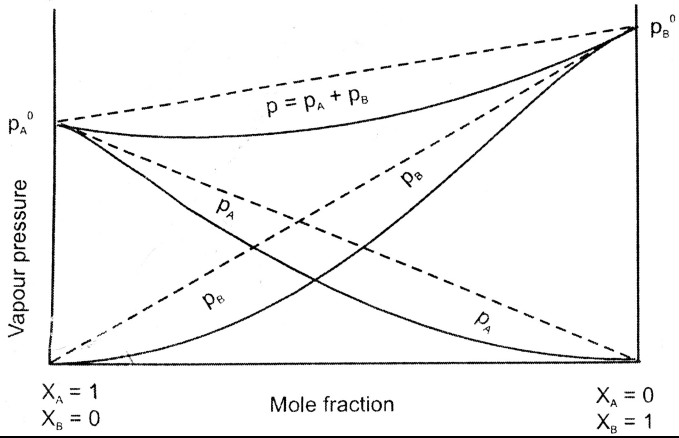
			SET B		
			SECTION A		
-	1	-	Same as Q. 5 of B1	1	1
-	2	-	Same as Q. 6 of B1	1	1
-	3	-	Same as Q. 8 of B1	1	1
-	4	-	Same as Q. 10 of B1	1	1
-	5	-	Same as Q. 12 of B1	1	1
-	6	-	Same as Q. 14 of B1	1	1
-	7	-	Same as Q. 9 of B1	1	1
-	8	-	Same as Q. 15 of B1	1	1
-	9	-	Same as Q. 13 of B1	1	1
-	10	-	Same as Q. 1 of B1	1	1
-	11	-	Same as Q. 2 of B1	1	1
-	12	-	Same as Q. 4 of B1	1	1
-	13	-	Same as Q. 3 of B1	1	1
-	14	-	Same as Q. 7 of B1	1	1
-	15	-	Same as Q. 11 of B1	1	1
-	16	-	D	1	1
-	17	-	Same as Q. 20 of B1	2	2
-	18	-	Same as Q. 21 of B1	2	2
-	19	-	Same as Q. 24 of B1	2	2
-	20	-	Same as Q. 17 of B1	2	2
-	21	-	Same as Q. 18 of B1	2	2
-	22	-	Same as Q. 28 of B1	2	2
-	23	-	Same as Q. 27 of B1	2	2
-	24	-	Same as Q. 19 of B1	2	2
-	25	-	Same as Q. 22 of B1	2	2
-	26	-	Same as Q. 23 of B1	2	2
-	27	-	Same as Q. 25 of B1	2	2
-	28	-	Same as Q. 26 of B1	2	2

			SECTION B		
-	29	-	Same as Q. 34 of B1	2	2
-	30	-	Aufban –The electron in an atom are filled in the order of their increasing energies. (i) An orbital with a lower value for (n+l) is filled first (ii) If the value of (n+l) first is same for two orbitals then the orbital with lower value of n will be filled first.	1 ½ ½	2
-	31	-	Same as Q. 29 of B1	2	2
-	32	-	Ist law: Energy can neither be created nor destroyed. The total energy of the universe or an isolated system is constant. $\Delta u = q + w$	2	2
-	33	-	$pH = - \log [H_3O^+]$ $5 = - \log[H_3O^+]$ $\text{Log } [H_3O^+] = -5$ Taking on 1 logs $[H_3O+] = 10^{-5}$ mol dem^{-3}.	½ ½ ½ ½	2
-	34	-	Same as Q. 31 of B1	2	2

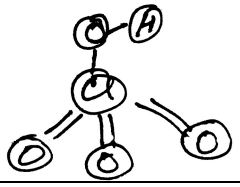
-	35	-	XeoF4 – shape – square pyramidal Hybridization sp^3d^2  	½ ½ 1	2
-	36	-	Same as Q. 37 of B1	2	2
-	37	-	Same as Q. 36 of B1	2	2

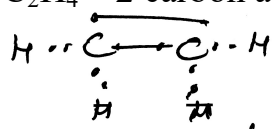
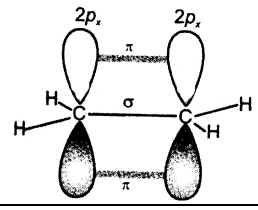
-	38	-	It is an electrostatic force of attraction that holds the cation and onion together in a compound. Formation of NaCl : Kossel's theory Na (11) = 2,8,1 Na (11) = 2,8,1 Na → $Na^+ + e^-$ 2,8,1 2,8 cl (17) = 2,8,7 cl + e^- → cl^- 2,8,7 2,8,8 Na cl → Na^+ cl^- 2,8,1 2,8,7 2,8 2,8,8	1 ½ ½ 1	3
-	39	-	(i) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O\ \Delta H = -890.2\ kJ/Mol$ (ii) $C + O_2 \rightarrow CO_2\ \ \Delta H = -393.4\ kJ/mol$ (iii) $H_2 + \frac{1}{2}O_2 \rightarrow H_2O\ \ \Delta H = -285.7\ kJ/mol$ Required equation $\underset{(graphin)}{C} + 2H_{2(g)} \rightarrow CH_{4(g)},\ \Delta H = ?$ Multiply eg (iii) by 2, add it to (ii) and then subtract from (i) $\Delta H = -393.4 + 2(-285.7) - (-890.2) kJ\ mol^{-1}$ $= -74.6\ kJ\ mol^{-1}$	1 1 1	3
-	40	-	Same as Q. 41 of B1	3	3
-	41	-	Same as Q. 40 of B1	3	3

-	42	-	Vapour pressure lowering happens when a non-volatile solute is added to an aqueous solution. The vapour phase of this solution contains molecules / vapours of solvent A only. $P_A = P_A^o \times A$ Relative lowering of V.P. for a solution is equal to the mole fraction of the solute, when only the solvent is volatile. Expression - $X_A + X_B = 1$ $P_A = P_A^o \times A$ $P_A = P_A^o \left(1 - X_B\right)$ $\frac{P_A}{P_A^o} = 1 - X_B$ Where M_A and W_A are molecules mass and mass in g of solvent $\frac{P_A^o - P_A}{P_A^o} = X_B = \frac{W_B \cdot M_A}{W_A \cdot M_B}$ M_B and W_B are molecular mass and mass in g of solute. $P_A^o - P_A = \text{lowering of v.p.}$ $\frac{P_A^o - P_A}{P_A^o} = \text{Relative lowering of v.p.}$ **OR**	1 1 ½ ½ 1 ½ ½	5

			Liquid pairs in which A-B molecular interactions are stronger than A-A or B-B molecular interactions show negative deviation. Example – Chloriferon + Acetone  Hydrogen bond function takes place so chloroferon-acetone interaction is stronger 	2 2 1	
-	43	-	Same as Q. 43 of B1	5	5

			SET C		
			SECTION A		
-	-	1	Same as Q. 5 of B1	1	1
-	-	2	Same as Q. 6 of B1	1	1
-	-	3	Same as Q. 16 of B1	1	1
-	-	4	Same as Q. 14 of B1	1	1
-	-	5	Same as Q. 12 of B1	1	1
-	-	6	Same as Q. 13 of B1	1	1
-	-	7	Same as Q. 1 of B1	1	1
-	-	8	Same as Q. 2 of B1	1	1
-	-	9	Same as Q. 3 of B1	1	1
-	-	10	Same as Q. 4 of B1	1	1
-	-	11	Same as Q. 7 of B1	1	1
-	-	12	Same as Q. 8 of B1	1	1
-	-	13	Same as Q. 9 of B1	1	1
-	-	14	Same as Q. 10 of B1	1	1
-	-	15	A	1	1
-	-	16	Same as Q. 11 of B1	1	1
-	-	17	Same as Q. 28 of B1	2	2
-	-	18	Same as Q. 27 of B1	2	2
-	-	19	Same as Q. 26 of B1	2	2
-	-	20	Same as Q. 25 of B1	2	2
-	-	21	Same as Q. 24 of B1	2	2
-	-	22	Same as Q. 17 of B1	2	2
-	-	23	Same as Q. 18 of B1	2	2
-	-	24	Same as Q. 19 of B1	2	2
-	-	25	Same as Q. 20 of B1	2	2
-	-	26	Same as Q. 21 of B1	2	2
-	-	27	Same as Q. 22 of B1	2	2
-	-	28	Same as Q. 23 of B1	2	2

			SECTION B		
-	-	29	Same as Q. 34 of B1	2	2
-	-	30	Pauli's exclusion principle – No two electrons can have all the four quantum numbers to be same. Example if for a given electron n=2, l=1, m=1 and m = + ½ then no other electron in the atom can have the same set of quantum number.	2	2
-	-	31	Same as Q. 31 of B1	2	2
-	-	32	$A_rH^o = \sum B.E.(\text{reactant bonds}) - \sum B.E.(\text{product bonds})$ $-46\ kJ\ mol^{-1} = 3\times 218\ kJ\ mol^{-1} + 973\ kJ\ mol^{-1} - B.E.(NH_3)$ $B.E.\ NH_3\ \text{bonds} = 1673\ kJ\ mol^{-1}$ $B.E\ (N-H) = 557.7\ kJ\ mol^{-1}$	½ ½ ½ ½	2
-	-	33	HCl completely ionizes as $HCl + H_2O \rightarrow H_3O^+ + Cl^-$ $\left[H_3O^+\right] = [HCl] = 0.03\ N\ (\text{given})$ $= 3\times 10^{-2} N = 3\times 10^{-2} m \begin{pmatrix} HCl \\ \text{Monobasic} \end{pmatrix}$ $pm = -\log\left[H_3D^+\right]$ $= -\log\left(3\times 10^{-2}\right)$ $= -\left(\log 3 + \log 10^{-2}\right)$ $= -(0.4771 - 2) = 1.5229$	½ ½ ½ ½	2
-	-	34	Same as Q. 29 of B1	2	2
-	-	35	HOCl < HOClO < $HOClO_2$ < $HOClO_3$ 	1×2	2
-	-	36	Same as Q. 37 of B1	2	2
-	-	37	Same as Q. 36 of B1	2	2

-	-	38	Ethene = C_2H_4 – 2 carbon atom  It has a double bond (one σ one π) π bond is formed by pure p-orbitals $C = 6 = 1s^2 2s^2 2p_x^1 2p_y^1 2p_z^0$ (ground state) $= 1s^2 \underbrace{2s^1 2p_x^1 2p_y^1}_{sp^2} 2p_z^1$  	1 1 1	3
-	-	39	Desired equation : $C_{(S)} + 2S_{(S)} \rightarrow CS_{2(l)} \quad \Delta H = ?$ Given : (i) $CS_2 + 3O_2 \rightarrow CO_2 + 2SO_2 \quad \Delta H = -110.2 \text{ kJ mol}^{-1}$ (ii) $S + O_2 \rightarrow SO_2 \quad \Delta H = -297.4 \text{ kJ mol}^{-1}$ (iii) $C + O_2 \rightarrow CO_2 \quad \Delta H = -394.5 \text{ kJ mol}^{-1}$ Eq. (iii) + 2 × Eq. (ii) – Eq. (i) $\Delta H = -394.5 + 2(-297.4) - (-110.2)$ $= -394.5 + (-594.8) - (-110.2)$ $= -879.1 \text{ kJ mol}^{-1}$	1 ½ ½ ½ ½	3
-	-	40	Same as Q. 41 of B1	3	3
-	-	41	Same as Q. 40 of B1	3	3

-	-	42	(a) Because it depends on the number of particle of a solute in solution $\frac{P_A^o - P_A}{P_A^o} = X_B = \frac{W_B \cdot M_A}{M_B \cdot W_A}$ (b) WA = 100g, WB = 1.0 g P^o = 195 mm, P_s = 192. 5 mm M_1 = 589 mol^{-1} $\frac{P^o - P_S}{P^o} = \frac{n_B}{n_A} = \frac{W_A / M_A}{M_A / W_A}$ $\frac{195 - 192.5}{195} = \frac{1.0 / M_B}{100 / 58} = \frac{58}{100\ M_B}$ $M_B = \frac{58}{100} \times \frac{195}{2.5} = 45.24\ g\ mol^{-1}$ **OR** (a) When a non-volatile solute is dissolved in a solvent to form a solution, its boiling point rises due to the non-formation of vapours by non-volatile solute. $\Delta T_b = k_b m$ (b) $\text{Molality of glucose} = \frac{\text{wt. of glucose}}{\text{mol. wt.}} \times \frac{1000}{\text{wt. of solvent}}$ $= \frac{0.52}{180} \times \frac{1000}{80.2} = 0.036\ m$ $\Delta T_b = k_b m = 0.52 \times 0.036$ $= 0.018\ k$ $\text{Boiling point} = 373 + 0.018$ $= 373.018$ $= 373.02\ k$	1 1 1 ½ ½ 1 1 1 ½ ½ ½ ½ 1	5 5
-	-	43	Same as Q. 43 of B1	5	5